

HoneyGotchi Pioneer Blueprint

The complete baseline: mission, parts, build steps, failed boots, SD-card lessons, network troubleshooting, Cybermap connection, and the road from a home lab to ground-to-space defense.

AI or not, I am the wizard.

Built by Jacob R. Zepeda
HoneyGotchi Baseline JRZ
September 2026

DOCUMENT RULE

This is not a polished myth written after success. It is the engineering record of what we tried, what failed, what each failure taught us, and what we do next. A pioneer keeps the map so the next crossing is wiser.

What HoneyGotchi is becoming

HoneyGotchi began as a physical cybersecurity companion: a small, friendly device that makes network awareness understandable. It is inspired by the spirit of portable security tools and virtual pets, but its purpose is our own - observe, learn, explain, and eventually help defend.

The baseline promise

- Run on affordable hardware we can assemble and understand ourselves.
- Teach Linux, networking, wireless behavior, logs, and defensive security through real use.
- Show status through a tiny screen and a simple personality instead of a wall of warnings.
- Treat privacy and authorization as design rules: observe our own environment, never trespass.
- Grow into a research platform for anomaly detection, UEBA, telemetry, and autonomous mitigation.

The long horizon

The home device is the proving ground. The deeper idea is a trustworthy defense companion that can watch small signals, remember normal behavior, explain anomalies, and recommend or perform tightly limited responses. That same pattern can later be studied for satellite ground systems and ground-to-space communications.

The name is a standard

HoneyGotchi is not valuable because it copies another device. It is valuable because it carries Jacob's method: build the machine, understand the machine, document the truth, and improve one layer at a time.

The baseline architecture

Layer	Purpose	First proof
Body	Raspberry Pi-class computer, storage, power, and enclosure.	The board powers reliably and reads a known-good flashed card.
Senses	Wi-Fi/network observations and later optional environmental sensors.	The device joins the authorized network and reports its own status.
Mind	Linux services, event parser, memory, scoring, and rules.	A local process classifies harmless demo events.
Face	Mini display, status icon, mood, and simple explanations.	Boot, network, quiet, curious, and alert states render clearly.
Map	Cyber Pulse / Cybermap web view.	Device state and clearly labeled demo signals appear without fake live claims.
Research	Anomaly detection and bounded response experiments.	Recorded baseline data supports repeatable lab tests.

Signal flow

AUTHORIZED INPUT -> NORMALIZE -> COMPARE TO BASELINE -> SCORE -> EXPLAIN -> DISPLAY -> LOG -> OPTIONAL HUMAN-APPROVED ACTION

The human stays in command. HoneyGotchi may suggest, visualize, and automate low-risk housekeeping, but higher-impact actions require explicit approval and a rollback path.

V1 success definition

- Stable boot from a verified microSD card.
- Reliable network connection with a discoverable hostname or IP.
- A local service starts automatically and writes readable logs.
- The screen shows a friendly state and at least one real device metric.
- Cybermap can show labeled demo data and, later, privacy-safe device telemetry.

Build inventory and bench rules

Item	Role	Check before use
Single-board computer	Runs the operating system and HoneyGotchi services.	Correct model, no visible damage, adequate cooling.
microSD card	Boot storage and operating system.	Known-good brand; verify capacity; keep a spare.
USB card reader	Lets the computer flash the microSD card.	Test with another card; readers can fail too.
Power supply	Provides stable voltage and current.	Use a board-appropriate supply, not a random low-output cable.
Mini display	HoneyGotchi's face and local status.	Confirm interface, voltage, pinout, and driver support.
Jumper wires / headers	Connect display and future sensors.	Correct connector gender and clean seating.
Case / standoffs	Protects the board from shorts and strain.	No metal contact under the board; airflow remains open.

Bench discipline

- Photograph wiring before powering on.
- Change one variable at a time.
- Label every card and reader.
- Keep a short test log: time, change, light behavior, network result.
- Never remove power while the card is being written.

What we deliberately did not assume

A green light does not automatically mean a successful boot. A card disappearing from Windows does not automatically mean it is dead. A device missing from the Wi-Fi list does not automatically mean the Wi-Fi password is wrong. Each symptom needs a controlled test.

The clean path from parts to first boot

#	Action	Proof before continuing
1	Inspect the board, card, reader, display, cables, and power supply.	No damage; card and reader appear on the computer.
2	Use the official imaging tool to select the correct OS and board.	Image choice matches the hardware.
3	Set hostname, user, Wi-Fi country, SSID, password, and SSH if needed.	Settings are reviewed before writing.
4	Write and verify the image to the microSD card.	Imager reports successful verification.
5	Safely eject the card, insert it fully into the board, then power on.	Power light is stable; activity light changes during boot when supported.
6	Wait several minutes for first boot.	Router/device list shows the hostname or a new IP.
7	Connect locally or by SSH and confirm OS, time, storage, and network.	Commands return expected values without I/O errors.
8	Update packages, then install one HoneyGotchi component at a time.	Reboot remains successful after each layer.
9	Wire and test the display using a minimal screen test.	Text or a test pattern appears before the full UI is installed.
10	Enable the service at startup and preserve logs.	A reboot returns HoneyGotchi to a known healthy state.

Golden image rule

Once the operating system, network, updates, and display test all work, create a backup image. That becomes the recoverable baseline before experimental security features are added.

What happened on the first crossing

Observed symptom	What it could mean	Lesson / next test
The device did not appear on Spectrum Wi-Fi.	Boot never completed, Wi-Fi settings were wrong, the wrong band/country was selected, or the router list lagged.	Prove boot first. Then inspect router clients, try hostname lookup, Ethernet if available, or a monitor/keyboard.
The green light stayed steady instead of blinking.	It may only indicate power; activity LED behavior differs by board and state.	Do not diagnose from color alone. Correlate LEDs with boot media, display output, network presence, and power quality.
The SD card looked empty or disappeared after flashing.	Linux partitions may be unreadable to Windows; Windows may show only a small boot partition or ask to format.	Do not format it just because Windows cannot read the Linux filesystem. Check Disk Management or the imaging tool.
A different card/reader behaved differently.	The original card, reader, adapter, USB port, or write process may be faulty.	Swap one component at a time. A known-good spare turns guessing into evidence.
The card no longer appeared at all.	Poor contact, failed reader, failed card, missing drive letter, or corrupted partition table.	Test the reader with a known-good card, the card in another reader, and inspect disk management before declaring it dead.
Repeated reflashing did not create confidence.	The process lacked a recorded verification checkpoint.	Capture the imager result and card identity; do not repeat the same uncertain action without a new test.

When HoneyGotchi does not appear

Start with the physical layer and climb upward. Never skip directly to software theories when power or storage has not been proven.

Decision	YES	NO
Does the computer detect the card before flashing?	Record size/brand and continue.	Try another port, reader, adapter, then card.
Does imaging finish with verification?	Safely eject and boot.	Stop. Re-seat, reformat through the imager, or replace suspect media.
Is power appropriate and stable?	Observe boot activity for several minutes.	Use the correct supply/cable before further diagnosis.
Does the board show real boot activity or display output?	Move to network checks.	Recheck board model, image, card seating, and boot logs.
Does the router show a new client/hostname?	Record the IP and connect.	Check SSID/password/country; test Ethernet or local console.
Can the device reach the gateway and internet?	Install the minimal HoneyGotchi service.	Test IP, gateway, DNS, and time separately.
Does the service survive reboot?	Freeze a baseline backup.	Inspect service status and logs; fix one failure at a time.

Useful evidence to capture

- Board model and power-supply rating.
- Card brand, capacity, and which reader/adapter was used.
- Imager success or error message.
- Exact LED pattern over the first 60 seconds.
- Router client list, hostname, IP, and timestamp.
- Any screen output, service status, and last 30 log lines.

From one companion to the wider field

Cybermap is the visual bridge between the small device and the big mission. It should help a person see how signals move without pretending that every dot is an attack.

Map layer	Today	Later
Demo signals	Clearly labeled simulated DNS, web, authentication, vulnerability, and alert events.	Scenario playback for training and portfolio demonstrations.
HoneyGotchi node	A card explaining the device's role, status, and current build stage.	Opt-in, privacy-safe telemetry from the local device.
Explanation	Observe -> correlate -> explain.	Confidence, source, baseline deviation, and recommended human action.
Defense horizon	Home-lab network view.	Ground-station, telemetry, communication-link, and satellite-defense research models.

Honesty rules

- Label simulated, public-source, and local-device data separately.
- Never imply a location is under attack without verified evidence.
- Collect the minimum needed; redact private addresses and identifiers.
- Show uncertainty and source timestamps.
- Make pause, filtering, and deletion understandable.

Website integration

The Cyber Pulse page now carries a Cybermap roadmap and a direct link to this field manual. Visitors can see both the living visualization and the engineering history behind it - the machine, the mistakes, and the next frontier.

What this project proves about the builder

Skill	Evidence created by the project
Linux administration	Imaging, boot diagnosis, services, packages, logs, storage, and recovery.
Networking	DHCP, DNS, Wi-Fi association, addressing, gateway tests, and traffic context.
Hardware troubleshooting	Power, media, readers, connectors, displays, controlled substitution, and fault isolation.
Cybersecurity operations	Event classification, baselines, anomaly thinking, alert honesty, and defensive boundaries.
Software engineering	Modular services, state handling, logging, testing, documentation, and versioned milestones.
AI research judgment	Explainability, confidence, safe autonomy, human approval, and measured claims.
Communication	A friendly interface and a public blueprint that translates technical work for ordinary people.

Portfolio proof package

- One-minute working demo video.
- Architecture diagram and threat model.
- Build log with failures and fixes.
- Source repository with setup instructions and screenshots.
- Cybermap demo with honest data labels.
- A short research note connecting the baseline to satellite telemetry security.

The honest truth

A half-working device with a rigorous troubleshooting record can prove more engineering ability than a copied project that boots once. The value is not that nothing broke. The value is that Jacob learned to interrogate the failure instead of surrendering to it.

The road ahead

Milestone	Deliverable	Gate
M0 - Recover boot	Verified card, stable power, local or network access.	Two clean reboots.
M1 - Give it a face	Display test plus five clear states.	Readable at arm's length.
M2 - Give it memory	Local event log and normal-behavior baseline.	Seven-day sample without data loss.
M3 - Give it judgment	Rule-based anomaly score with explanations.	Known tests produce repeatable results.
M4 - Connect Cybermap	Opt-in status and sanitized event summaries.	No private identifiers leave the device.
M5 - Controlled response	Human-approved, reversible low-risk actions.	Every action logged and undoable.
M6 - Research bridge	Ground-to-space lab scenario and evaluation plan.	Threat model, metrics, and limits documented.

Immediate next move

Do not add more features until M0 is proven. Bring one known-good card, one known-good reader, the correct power supply, and a written Wi-Fi configuration to the bench. Flash once, verify once, boot once, and collect evidence. The wizard's power is disciplined observation.

THE PIONEER'S CREED

We build before the road is clear.

We do not hide the broken cards, silent lights, missing devices, or wrong turns. Those are not stains on the work. They are marks on the map.

We respect the machine enough to learn what it is telling us. We change one thing at a time. We record the truth. We protect what is ours and never trespass into what is not.

We use AI as a forge, a compass, and a second pair of eyes - never as a substitute for judgment, courage, or responsibility.

We are pioneers of our own lives. We do not wait for permission to become builders. We earn skill through repetition, humility, and the refusal to quit when the first boot fails.

AI or not, I am the wizard.

BUILDINGWITHJACOB

Engineering in public. One working layer at a time.